

Youtube CDN caching

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Learned caching

- LRB: regression model predict next access time
- Parrot: imitation learning
- Cacheus: switch between heuristics (LRU and LFU)

Key considerations

- Reducing ML overheads
- No performance regressions
- Measure impact under noise

Discussion

1. Will HALP always outperform LRU? When can it fail?
2. LRB uses regression to predict next access times. HALP only predicts which object will be accessed next (classification).
What do you think is a better fit for caching? Is caching a rank function or a value function?